

TSHELO

Test Partner Safeguards

Bank of Botswana Regulatory Sandbox Application

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This document sets out the safeguards Data Sentinels has in place to manage risks arising from its reliance on third-party partners during the Bank of Botswana Regulatory Sandbox testing period (May 2026 to October 2026). It covers the nature of each partner relationship, the risks associated with that reliance, and the specific safeguards and contingency measures in place for each.

1. Fundamental Risk Reduction Principle

Tshelo's architecture is deliberately designed to minimise dependency on third-party partners in a way that could affect customer funds or data. The key principle is:

Tshelo does not hold, process, or transmit any customer funds at any time.

This means that a failure by any mobile money partner does not result in any loss of customer funds attributable to Tshelo. The only risk from partner failure is loss of a convenience feature (SMS detection). The fallback of manual contribution entry is always available and requires no third-party dependency.

2. Partner Register and Risk Assessment

The following partners are relied upon during the sandbox testing period. Each is assessed for dependency level, risk, and safeguards.

2.1 Supabase (Infrastructure and Database Provider)

Element	Detail
Nature of relationship	Cloud infrastructure provider. Tshelo's backend database, authentication, and file storage run on Supabase, which is hosted on AWS.
Regulatory status	Supabase is a commercial cloud provider subject to AWS terms and SOC 2 Type II certification. Not a regulated financial institution. No financial data or funds are processed by Supabase.
Dependency level	HIGH. All platform data and user authentication rely on Supabase.
Primary risk	Extended platform outage if Supabase or AWS infrastructure experiences a significant failure.
Status	ACTIVE

Safeguards in place:

- Supabase provides a 99.9% uptime SLA backed by AWS infrastructure, equating to less than 9 hours downtime per year

- Automated database backups run every 24 hours. Maximum data loss in any outage is 24 hours of records
- All user data is encrypted at rest (AES-256) and in transit (TLS 1.2+) by Supabase
- Row Level Security enforced at database level ensures no cross-user data access even in the event of an application error
- PDF export feature allows users to save their fund records independently of platform availability at any time
- If Supabase were to discontinue service, all data can be exported and migrated to an alternative PostgreSQL provider within the recovery time objective

2.2 Orange Money, MyZaka (Mascom), and Smega (BTC)

Element	Detail
Nature of relationship	Tshelo reads SMS notifications from these providers on the user's device to detect contribution amounts. Tshelo does not have a formal API agreement with any provider during the sandbox period. SMS detection is device-level with user permission only.
Regulatory status	All three are licensed electronic payment service providers regulated by the Bank of Botswana. They have their own KYC, AML, and consumer protection obligations. BTC has expressed support for the Tshelo product and API discussions are in progress.
Dependency level	MEDIUM. SMS detection is a core convenience feature but manual contribution entry is always available as a fallback requiring no mobile money provider dependency.
Primary risks	(1) Change in SMS format that breaks detection. (2) Mobile money provider restricts SMS access. (3) Provider experiences an outage affecting users ability to send or receive money.
Status	IN DISCUSSIONS

Safeguards in place:

- Manual contribution entry is always available as a fallback. If SMS detection fails entirely, the platform continues to function without interruption
- SMS parsing is designed to be flexible across format variations. If a provider changes their SMS format, an update can be deployed within 48 hours
- Tshelo does not rely on mobile money providers for fund storage, payment processing, or user KYC. The dependency is limited to the convenience of automatic SMS parsing
- In the event of a mobile money provider outage, fund organisers are advised to record contributions manually until the provider service is restored
- BTC API integration discussions are ongoing. A formal API agreement will be executed before any direct API connection is established and will be provided to BoB as a supplementary attachment
- All three providers are licensed and regulated by the Bank of Botswana, providing an additional layer of regulatory oversight over these partners

2.3 Expo and React Native (Mobile App Framework)

Element	Detail
Nature of relationship	Expo is the framework used to build and distribute the Tshelo mobile app. React Native is the underlying cross-platform mobile development library.
Regulatory status	Not a regulated entity. Open source framework with a commercial hosting layer (Expo EAS). No customer data or funds pass through Expo.
Dependency level	MEDIUM. The app is built on Expo but can be ejected to a standalone React Native build if Expo discontinues service.
Primary risk	Expo discontinues a service or introduces a breaking change that requires urgent app update.
Status	ACTIVE

Safeguards in place:

- Expo is a widely adopted, stable framework used by thousands of production apps globally. Discontinuation risk is very low
- The app can be ejected from the Expo managed workflow to a bare React Native build if needed, without loss of functionality
- App store distribution (Google Play and Apple App Store) is independent of Expo's managed services

2.4 Google Play Store and Apple App Store

Element	Detail
Nature of relationship	Tshelo is distributed through the Google Play Store and Apple App Store. Users download the app from these stores.
Dependency level	HIGH for new user acquisition. LOW for existing users who have already installed the app.
Primary risk	App removed from one or both stores due to policy violation, preventing new downloads during the sandbox period.
Status	ACTIVE

Safeguards in place:

- App content and functionality will be reviewed against both store policies before sandbox launch to ensure full compliance
- Android APK direct distribution is available as a fallback for Botswana users if the Google Play Store listing is unavailable
- Existing users who have installed the app are unaffected by any store listing issue as the app continues to function
- The Bank of Botswana will be notified within 24 hours if either app store listing is suspended

3. Partner Risk Summary Table

Partner	Dependency	Risk to Funds	Risk to Data	Primary Safeguard
Supabase	HIGH	NONE	MEDIUM	99.9% SLA. 24hr backups. PDF export fallback.
Orange Money, MyZaka, Smega	MEDIUM	NONE	LOW	Manual entry fallback. BoB-licensed providers.
Expo and React Native	MEDIUM	NONE	LOW	Ejection to bare React Native available.
Google Play and Apple App Store	HIGH	NONE	NONE	APK direct distribution fallback. Existing users unaffected.

4. Partnership Agreement Status and Plan

The following table sets out the current status of formal agreements with each partner and the plan for formalisation during the sandbox period:

Partner	Status	Nature of Agreement Planned	Timeline
Supabase	ACTIVE	Standard commercial subscription agreement. Terms of Service accepted.	In place. No further action required.
BTC (Smega)	IN DISCUSSIONS	API access agreement for SMS notification integration.	Formal agreement to be executed during Month 1 of sandbox (May 2026). To be provided to BoB as supplementary attachment.
Orange Money	PLANNED	API or data sharing agreement for SMS notification integration.	Engagement to be initiated in Month 1 of sandbox. Agreement targeted for Month 2 (June 2026).
MyZaka (Mascom)	PLANNED	API or data sharing agreement for SMS notification integration.	Engagement to be initiated in Month 1 of sandbox. Agreement targeted for Month 2 (June 2026).
Expo	ACTIVE	Standard commercial subscription. Expo EAS Terms of Service accepted.	In place. No further action required.
Google Play and Apple App Store	PLANNED	Developer account agreements. App store listing compliance review.	Developer accounts to be established and app submitted for review in Month 1 of sandbox (May 2026).

5. Data Sentinels' Commitments Regarding Partner Management

- Data Sentinels will not enter into any new partner dependency that could affect customer funds or data without first notifying the Bank of Botswana
- All formal API agreements executed with mobile money providers during the sandbox period will be provided to the Bank of Botswana as supplementary attachments within 5 business days of execution
- Partner performance will be monitored throughout the sandbox period. Any partner failure that affects platform availability or user data will be reported to the Bank of Botswana within 48 hours
- A partner review is included in every monthly sandbox progress report submitted to the Bank of Botswana
- Data Sentinels will not engage with any partner that is not compliant with applicable Botswana law or that poses an unacceptable risk to user data or platform integrity